
The Simplex Method and Diameter of Polyhedra

In Lecture 1 we studied linear programming (LP) from a structural perspective, including facial structure, representation theorems and duality theory. This lecture covers the most popular method for solving LPs, the *simplex method*. Before explaining it in detail, we begin in Section 1 with a brief overview of the main classes of LP algorithms, explaining what it means to solve an LP approximately and highlighting the major open problem of solving LP in *strongly* polynomial time.

Dantzig’s simplex method [Dan51] solves an LP by walking from vertex to vertex along *improving edges* until it reaches an optimal vertex (or unbounded ray). The simplex method exhibits a striking gap between its practical and theoretical worst-case performance: decades of computational experience suggest that the number of steps it takes is only *linear* in the number of constraints [Sha87], whereas there are LP instances for which all known implementations require an exponential number of steps [KM72]. Since the simplex method traverses the graph of vertices and edges of a polyhedron, the diameter of this graph — the length of the longest shortest path between two vertices — lower bounds the worst-case performance of simplex. An exponential diameter upper bound was shown by Barnette and Larman independently [Lar70, Bar74], and a quasi-polynomial upper bound was given by Kalai and Kleitman [KK92], while the best known lower bound constructions remain linear [San12]. A major open problem is whether the diameter is in fact polynomial in the number of constraints, known as the *polynomial Hirsch Conjecture*.

In Section 2, we explain the basic geometry of the simplex method, why it works, and how to implement the abstract algorithm in standard form. In Section 3, we survey popular rules for selecting the improving edge, known as *pivot rules*, and present the exponential lower bound for a worst-case pivot rule via the *Klee-Minty* cube [KM72]. Finally, in Section 4, we give a modern treatment of the Barnette-Larman and Kalai-Kleitman diameter upper bounds using the abstraction of connected layer families (CLF), introduced by Eisenbrand, Hähnle, Razborov and Rothvoß [EHRR10].

1 The Landscape of LP Algorithms

We begin with an overview of algorithmic approaches to linear programming and fix the vocabulary used to compare them. The algorithms below are naturally stated in different formulations of LP, so we will consider them in either the *standard form*

$$\min c^\top x \text{ s.t. } \mathbf{A}x = b, \ x \geq \mathbf{0}, \quad \text{or the geometric form} \quad \max c^\top x \text{ s.t. } \mathbf{A}x \leq b,$$

with $\mathbf{A} \in \mathbb{R}^{m \times n}$ in both cases: simplex and interior point methods are most often stated in standard form, the ellipsoid method in geometric form (the two forms are interchangeable, though with different parameters — see Section 2.3). In either form, the triple $(\mathbf{A}, b, c)$ captures the input data, and we write L for its total *bit encoding length* when the entries are rational.

Families of algorithms. Four broad families of LP algorithms have emerged over the past eighty years (Figure 1). *Simplex methods* [Dan51] walk from vertex to vertex along edges of the feasible region, improving the objective at every step. *Interior point methods* [Kar84] instead traverse the strict interior of the feasible region, following a smooth trajectory (the *central path*, the subject of Lectures 4 and 5) that leads to an optimal solution. *Cutting plane methods*, most notably the

ellipsoid method [Kha79], maintain an ellipsoid guaranteed to contain the optimal solutions: at every step, the current ellipsoid is cut in half through its center — along a constraint violated by the center, or along the objective when the center is feasible — and the half that still contains the optimal solutions is enclosed in a new ellipsoid whose volume is smaller by a fixed factor. *First-order methods* apply gradient-type updates to a suitable reformulation of the LP; they perform only matrix-vector products per iteration, and are attractive when $\mathbf{A}$ is very large and sparse, though their iteration counts scale polynomially in the desired accuracy rather than logarithmically. All four families are *iterative*, and, in each, the work per iteration is a fixed number of basic linear-algebraic operations — solving a linear system, computing a matrix-vector product, updating a matrix factorization. Comparing them therefore reduces, to first order, to comparing their *iteration counts*.

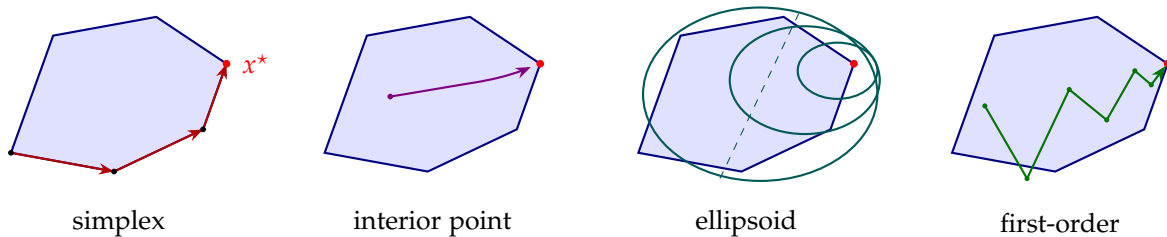


Figure 1: The four families at a glance, on the same feasible region with optimal solution x^* (red). Simplex walks from vertex to vertex along the boundary. Interior point methods follow a smooth curve through the interior. The ellipsoid method encloses the optimal solutions in ellipsoids of shrinking volume: each step cuts the current ellipsoid in half through its center (dashed line) and encloses the surviving half in a smaller ellipsoid. First-order methods take many cheap local steps, which need not stay feasible.

Exact versus approximate solvers. A fundamental division concerns what an algorithm produces. An *exact* solver terminates with an optimal solution — typically an optimal vertex or basis, together with a certificate of optimality in the sense of Lecture 1. The simplex method is exact by design: it stops at a vertex satisfying the local (hence global) optimality condition, after finitely many pivots. For an exact solver the natural complexity question is:

How many iterations are needed to reach an optimal solution?

An *approximate* solver instead drives down an error measure — ideally the *optimality gap* $g(x) := c^T x - \text{OPT}$ of the current feasible iterate, or, when the iterates are not kept feasible, a combination of the optimality gap and a *feasibility gap* measuring constraint violation. The gap itself is not observable (OPT is unknown!), so what each method actually reduces is a computable *proxy* for it: for interior point methods, the *barrier parameter* $\mu > 0$ indexing the current point on the central path, which bounds the duality gap of the associated primal–dual pair (Lectures 4 and 5); for the ellipsoid method, the *volume* of the enclosing ellipsoid, small volume forcing the remaining near-optimal solutions to have small gap; for first-order methods, typically a potential combining the duality gap of a primal–dual iterate pair with its primal and dual infeasibility (a *KKT residual*). These methods decrease their proxy at a predictable rate, without necessarily ever reaching optimality, and the natural question becomes:

How many iterations are needed to shrink the error by a factor of 2?

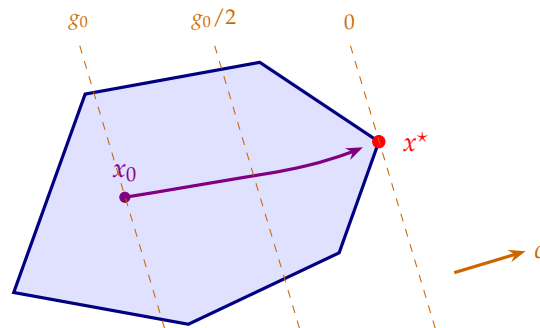


Figure 2: An approximate solver (here an interior point method, violet): starting at x_0 , the iterates converge to the optimum x^* . The dashed lines are level sets of the optimality gap: gap g_0 at the start x_0 , gap 0 at x^* , and the halfway line at gap $g_0/2$. Shrinking the gap by a factor of 2 means advancing from the g_0 -line to the $g_0/2$ -line.

From approximate to exact: bit complexity. Over the rationals, an approximate solver can be converted into an exact one: once the optimality gap is smaller than $2^{-\Theta(L)}$, the optimal vertex is the unique vertex near the current iterate, and can be recovered by rounding. The threshold $2^{-\Theta(L)}$ is where the bit size of the input enters the complexity bounds: an algorithm that halves the gap every T iterations solves the LP exactly in $O(T \cdot L)$ iterations. Algorithms of this kind are called *weakly polynomial*: the number of arithmetic operations is polynomial in L , depending on the bit sizes of the data and not merely on the dimensions m and n .

Strongly polynomial algorithms. It is natural to ask for more: an algorithm whose *combinatorial* work is independent of the precision of the numbers. This is formalized as follows [Meg83].

DEFINITION 1 (STRONGLY POLYNOMIAL ALGORITHM) *An exact LP algorithm is strongly polynomial if (i) it performs $\text{poly}(m, n)$ basic arithmetic operations ($+$, $-$, $\times$, $/$, comparisons), and (ii) it runs in space polynomial in the input size L (the numbers appearing in intermediate computations stay of polynomial bit length).*

Condition (i) says the *number of iterations* may depend only on the combinatorial size of the LP, not on the magnitudes of the data; condition (ii) rules out abusing exact arithmetic to compress super-polynomial computations into few operations on enormous numbers. Whether linear programming admits a strongly polynomial algorithm is among the most important open problems in optimization — Smale lists it as one of the mathematical problems for the twenty-first century [Sma98].

LP classes solvable in strongly polynomial time. A number of important LP classes are known to admit strongly polynomial algorithms. On the combinatorial side, these include maximum flow [EK72] and, by a celebrated result of Tardos, minimum-cost flow [Tar85]; more generally, Tardos showed that LPs whose constraint matrix $\mathbf{A}$ has polynomially bounded integer entries can be solved in time polynomial in m, n — independent of b and c [Tar86]. On the interior point side, Vavasis and Ye [VY96] gave an algorithm whose iteration count depends only on the constraint matrix $\mathbf{A}$ — through a condition measure essentially equivalent to the *circuit imbalance measure* $\kappa_{\mathbf{A}}$ of the next lecture [DHN20] — yielding strongly polynomial algorithms whenever

κ_A is bounded (for example, for minimum cost flow). A recent result of Dadush, Koh, Natar, Olver and Véghe gives a strongly polynomial interior point method for LPs whose constraint matrix has at most two nonzero entries per column [DKN⁺24]. Against these positive results stand lower bounds: Allamigeon, Benchimol, Gaubert and Joswig proved that no interior point method following the central path can be strongly polynomial [ABGJ18]. We will cover these interior point based approaches in Lectures 4 and 5.

2 The Simplex Method

Throughout this section, $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ with $A \in \mathbb{R}^{m \times n}$ of full column rank, and we consider the LP $\max \{c^\top x : x \in P\}$.

2.1 The vertex-edge graph, and the walk

The combinatorial object underlying the simplex method is the *vertex-edge graph* (or *1-skeleton*) of P : the graph $G(P) = (V(P), E(P))$ whose nodes are the vertices of P and whose (undirected) edges connect $u, v \in V(P)$ whenever $\text{conv}(u, v) = [u, v]$ is a bounded edge (one-dimensional face) of P . For a polygon, $G(P)$ is the cycle of corners; for the cube $[0, 1]^3$, the familiar cube graph with 8 nodes and 12 edges.

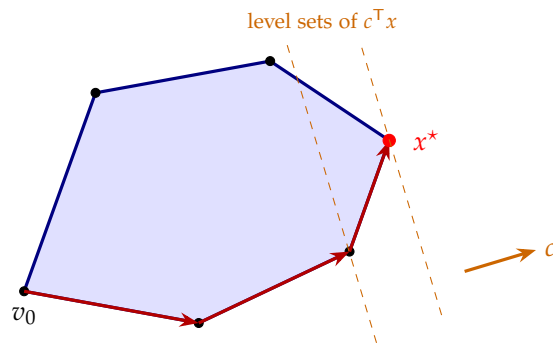


Figure 3: A simplex run on a polygon: starting at v_0 , repeatedly cross an edge that improves the objective, until no improving edge remains. The dashed lines are level sets $\{x : c^\top x = \text{const}\}$ of the objective, drawn through the last two vertices of the walk; the optimal vertex x^* is the point of P on the highest level set.

The simplex method implements the procedure suggested by Figure 3:

Abstract simplex method. Start at a vertex v of P . While some edge of P incident to v improves the objective: move along it, either to its other endpoint (a better vertex) or, if the edge is an unbounded ray, report *unbounded* and stop. When no incident edge improves, output v .

Two things need justification: that a non-optimal vertex always has an improving incident edge (correctness), and that the walk terminates. The rule by which one *chooses* among several improving edges is called the *pivot rule* — Dantzig’s rule, steepest edge, greatest improvement, shadow vertex, ... — and is the subject of Section 3; correctness holds for every pivot rule.

2.2 Correctness: the tangent cone

To reason about single vertices we introduce a local object: the cone of feasible movement directions.

DEFINITION 2 (TANGENT CONE) For $x \in P$, let $I(x) = \{i : a_i^\top x = b_i\}$ denote its tight constraints. The tangent cone of P at x is

$$T_P(x) := \{w \in \mathbb{R}^n : x + \varepsilon w \in P \text{ for some } \varepsilon > 0\} = \{w \in \mathbb{R}^n : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}.$$

(For the second equality: tight constraints must not increase, which is necessary and, for small ε , sufficient, since slack constraints tolerate small movements.)

The key local-to-global property is that, seen from any $x \in P$, the entire polyhedron lies within the tangent cone: $P \subseteq x + T_P(x)$, since $\mathbf{A}_I(y - x) = \mathbf{A}_I y - b_I \leq \mathbf{0}$ for every $y \in P$. At a vertex, the tangent cone is generated by the incident edge directions — this is a local application of Minkowski's theorem.

LEMMA 3 (EDGES GENERATE THE TANGENT CONE) Let $v \in V(P)$, and let E_v denote the set of edges of P incident to v (one-dimensional faces containing v , here including unbounded ones). Then $T_P(v)$ is a pointed cone with

$$T_P(v) = \text{cone}(e - v : e \in E_v), \quad \text{where } e - v := \{x - v : x \in e\}.$$

In words: the tangent cone at v is generated by the directions of the edges of P emanating from v ; its extreme rays are exactly these directions.

PROOF: Write $I = I(v)$, so $\text{rank}(\mathbf{A}_I) = n$ and $T := T_P(v) = \{w : \mathbf{A}_I w \leq \mathbf{0}\}$ is pointed. By Minkowski's theorem (Lecture 1) applied to T , we have $T = \text{cone}(R(T))$, the cone of its extreme ray generators, so it suffices to match extreme rays with incident edges. Let $[0, \infty)r$ be an edge of T , with tight set $J \subseteq I$, $\text{rank}(\mathbf{A}_J) = n - 1$, $\ker(\mathbf{A}_J) = \text{span}(r)$. Consider the face $F := \{x \in P : \mathbf{A}_J x = b_J\}$ of P . It contains v , and for small $\varepsilon > 0$ it contains $v + \varepsilon r$: constraints in J stay tight, constraints in $I \setminus J$ have $a_i^\top r \leq 0$, and constraints outside I are slack at v . Since F lies in the line $\{x : \mathbf{A}_J x = b_J\}$, it is an edge of P through v with direction r . Conversely, if E is an edge of P containing v with tight set $J' := \{i : a_i^\top x = b_i \ \forall x \in E\}$ and direction r pointing from v into E , then $\text{rank}(\mathbf{A}_{J'}) = n - 1$ (Lecture 1), $J' \subseteq I$, and $\{w \in T : \mathbf{A}_{J'} w = \mathbf{0}\} = \ker(\mathbf{A}_{J'}) \cap T = [0, \infty)r$ is an edge of T . $\square$

THEOREM 4 (CORRECTNESS OF THE SIMPLEX METHOD) Let $v \in V(P)$. Exactly one of the following holds:

- (a) $c^\top r \leq 0$ for every incident edge direction r at v , and then v is optimal;
- (b) some incident edge direction r has $c^\top r > 0$, and then moving along the corresponding edge either reaches a vertex of strictly larger objective value or certifies that the LP is unbounded.

PROOF: Suppose (a). For any $y \in P$ we have $y - v \in T_P(v) = \text{cone}(\text{edge directions})$ by Theorem 3, say $y - v = \sum_r \mu_r r$ with $\mu \geq \mathbf{0}$; then $c^\top y - c^\top v = \sum_r \mu_r c^\top r \leq 0$, so v is optimal. Otherwise some incident edge E has direction r with $c^\top r > 0$; along E the objective strictly increases, and E either

ends at a neighboring vertex of strictly better value, or is unbounded, in which case $r \in \text{rec}(P)$ with $c^\top r > 0$ certifies $\text{OPT} = +\infty$ (Lecture 1). $\square$

Note the pleasing symmetry with Lecture 1: there, we certified *global* optimality by writing c as a nonnegative combination of the tight constraint normals; here, we certify *local* optimality (which is global, by convexity!) by testing c against the tangent cone — the two tests are dual descriptions of the same geometry.

COROLLARY 5 (TERMINATION) *If every pivot strictly increases the objective — e.g., whenever no vertex has more than n tight constraints — then the simplex method terminates after at most $|V(P)| \leq \binom{m}{n}$ pivots, for any pivot rule.*

PROOF: The objective strictly increases, so no vertex is ever revisited; and $|V(P)| \leq \binom{m}{n}$, each vertex being determined by a choice of n tight constraints. $\square$

When vertices with more than n tight constraints exist (*degeneracy*), a pivot may keep the same vertex while exchanging tight constraints, and careless pivot rules can *cycle* forever. Anti-cycling rules exist — the simplest is Bland’s smallest-index rule [Bla77] — but we will assume non-degeneracy today and not dwell on this. The assumption is mild: as the following exercise shows, degeneracy can always be removed by an arbitrarily small perturbation of b , at no loss in the computed optimum.

Exercise 1 (Removing degeneracy by perturbation) Assume that $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ is pointed, i.e., $\text{rank}(A) = n$, and that the LP $\max\{c^\top x : x \in P\}$ is feasible and bounded. For $\varepsilon > 0$, let $\varepsilon^{(m)} := (\varepsilon, \varepsilon^2, \dots, \varepsilon^m)^\top$ and consider the perturbed polyhedron $P_\varepsilon := \{x \in \mathbb{R}^n : Ax \leq b + \varepsilon^{(m)}\}$.

1. Show that there is $\varepsilon_0 > 0$ such that for all $0 < \varepsilon < \varepsilon_0$, every vertex of P_ε has exactly n tight constraints.
2. Prove that there exists $J \subseteq [m]$, indexing n linearly independent rows, such that $v_\varepsilon := A_J^{-1}(b_J + \varepsilon_J^{(m)})$ is an optimal vertex of $\max\{c^\top x : x \in P_\varepsilon\}$ for arbitrarily small $\varepsilon > 0$. Show that, for any such J , the point $v_0 := A_J^{-1}b_J$ is an optimal vertex of the original LP.

In particular, running the simplex method on the perturbed LP — where no cycling can occur — recovers an optimal tight-row set, and hence an optimal vertex, of the original LP.

In practice one need not commit to a numerical value of ε : the perturbation can be implemented *symbolically*, comparing slacks — which are polynomials in ε — by the lexicographic order of their coefficients, i.e., by their behavior as $\varepsilon \rightarrow 0^+$. This is the classical *lexicographic pivoting rule*, one of the standard anti-cycling safeguards.

A pleasant byproduct of correctness is the connectivity of the vertex-edge graph — the property underlying the diameter question of Section 4.

COROLLARY 6 ($G(P)$ IS CONNECTED) *If P has a vertex, then $G(P)$ is connected: any two vertices of P are joined by a path of bounded edges.*

PROOF: Fix $u, v \in V(P)$. The trick is to choose an objective tailor-made so that v is its unique maximizer, and then let simplex find the path for us: take $c := \sum_{i \in I(v)} a_i$, the sum of the constraint normals tight at v . For every $y \in P$,

$$c^\top y = \sum_{i \in I(v)} a_i^\top y \leq \sum_{i \in I(v)} b_i = c^\top v,$$

with equality iff every constraint of $I(v)$ is tight at y , which forces $y = v$ since $\text{rank}(\mathbf{A}_{I(v)}) = n$. So v is the unique maximizer of c over P . Now run the abstract simplex method with objective c from u . Since $\max_{x \in P} c^T x = c^T v$ is finite, no improving edge is an unbounded ray, so by Theorem 4 every step crosses a bounded edge to a vertex of strictly larger c -value. No vertex is ever revisited and $|V(P)|$ is finite, so the walk stops at a vertex with no improving edge, which by Theorem 4(a) is optimal — i.e., it stops at v . The vertices visited form the required u - v path in $G(P)$. $\square$

2.3 The simplex method in standard form

So far we have worked in the *geometric form* $P = \{x : \mathbf{A}x \leq b\}$ of Lecture 1 — the right language for pictures: halfspaces, facets, tangent cones. It is not, however, the most convenient form for *computation*. To implement the abstract simplex method we switch to the *standard form*

$$\min c^T x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0},$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ has $m \leq n$ rows and full row rank. This is the right language for algebra — vertices become invertible column submatrices (*bases*), moving along an edge becomes solving a linear system (a *pivot*) — and it is the form in which solvers, and the next lecture's kernel-based condition measures, operate. The switch is local: only this subsection works exclusively in standard form; elsewhere we use whichever form is most convenient. Since the two forms describe the same class of objects, it is worth recording the translation precisely.

Standard form is itself geometric: the feasible set is the intersection of the affine subspace $\{\mathbf{A}x = b\}$ with the orthant $\{x \geq \mathbf{0}\}$, so all of Lecture 1 applies to it, with the n coordinate bounds $x_j \geq 0$ as the only inequality constraints. Conversely, a Lecture-1 polyhedron $P = \{x : \mathbf{A}x \leq b\}$ with $\text{rank}(\mathbf{A}) = n$ is mapped by the *slack coordinates* $s := b - \mathbf{A}x$ bijectively (and affinely) onto

$$\{s \geq \mathbf{0} : s \in b + \text{im}(-\mathbf{A})\},$$

an affine slice of the orthant $\mathbb{R}_{\geq 0}^m$ — i.e., a standard-form feasible set — and the i -th constraint of P is tight exactly when the coordinate s_i vanishes. The dictionary reads:

geometric form $\{x : \mathbf{A}x \leq b\}$	standard form $\{x : \mathbf{A}x = b, x \geq \mathbf{0}\}$
tight constraint $a_i^T x = b_i$	zero coordinate $x_j = 0$
vertex = n independent tight constraints	basic feasible solution: $x_N = \mathbf{0}$, $\mathbf{A}_B$ invertible
edge = $n - 1$ independent tight constraints	edge: $x_{[n] \setminus S} = \mathbf{0}$, $\ker(\mathbf{A}_S)$ one-dimensional ($ S = m + 1$)
tangent cone $\{w : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}$	$\{w \in \ker(\mathbf{A}) : w_j \geq 0 \text{ whenever } x_j = 0\}$
non-degenerate: no vertex with $> n$ tight	non-degenerate: $x_B > \mathbf{0}$ at every BFS

A remark on conventions: we keep the convention that m counts rows and n counts columns in *both* forms; beware that the number of *inequalities* is m in the geometric form but n in standard form, so statements like the Hirsch conjecture (“ m facets”) and basis counts (“ $\binom{n}{m}$ bases”) refer to different parameters.

REMARK 1 (VARIABLE BOUNDS) Capacitated variables, $\mathbf{Ax} = b$, $\mathbf{0} \leq x \leq u$, reduce to standard form by pairing each variable with a complementary slack variable: $x_j + \bar{x}_j = u_j$, $x, \bar{x} \geq \mathbf{0}$. The kernel of the enlarged matrix is $\{(w, -w) : w \in \ker(\mathbf{A})\}$ — a mirrored copy of $\ker(\mathbf{A})$ — so every kernel-based quantity (edge directions, and the circuit imbalance measure of the next lecture) carries over unchanged, while each bound $x_j \leq u_j$ becomes the nonnegativity of the complementary variable $\bar{x}_j$. This reduction will be used throughout the next lecture, where the complement of a flow variable is precisely its reverse residual arc.

DEFINITION 7 (BASIC FEASIBLE SOLUTIONS) A basis is a set $B \subseteq [n]$, $|B| = m$, with $\mathbf{A}_B$ (the corresponding columns) invertible. Its basic solution is given by $x_B = \mathbf{A}_B^{-1}b$ on the coordinates in B and $x_N = \mathbf{0}$ on $N := [n] \setminus B$; it is a basic feasible solution (BFS) if $x_B \geq \mathbf{0}$. The BFS is non-degenerate if $x_B > \mathbf{0}$.

BFSs are exactly the vertices of the feasible region (matching tight constraints: $\mathbf{Ax} = b$ plus $x_N = \mathbf{0}$ gives n tight, independent constraints), and, under non-degeneracy, two bases whose sets differ in one element correspond to vertices joined by an edge.

Edge directions and reduced costs. At a vertex, the local geometry of Theorem 3 — the incident edge directions and their objective slopes — takes a fully explicit form. Let x be a non-degenerate BFS with basis B , so that its tight constraints are the m equations $\mathbf{Ax} = b$ together with the $n - m$ bounds $x_j = 0$, $j \in N$ (non-degeneracy: $x_B > \mathbf{0}$, so no bound indexed by B is tight). An incident edge is obtained by relaxing *one* tight inequality $x_j = 0$, $j \in N$, while keeping the other $n - 1$ tight constraints tight (an edge is the face of $n - 1$ independent tight constraints). Algebraically, the corresponding direction $w^{(j)}$ solves

$$\mathbf{A}w^{(j)} = \mathbf{0}, \quad w_{N \setminus \{j\}}^{(j)} = \mathbf{0}, \quad w_j^{(j)} = 1 \quad (\text{normalization}),$$

which gives $w_B^{(j)} = -\mathbf{A}_B^{-1}\mathbf{A}e_j$ — this is exactly the direction computed in line 4 of Algorithm 1 below. In words: as the *entering* variable x_j increases by t , the basic variables must adjust to $x_B - t\mathbf{A}_B^{-1}\mathbf{A}e_j$ in order to preserve $\mathbf{Ax} = b$. There are exactly $n - m$ such directions, one per non-basic index; in the dictionary above, they are the extreme rays of the tangent cone $\{w \in \ker(\mathbf{A}) : w_N \geq \mathbf{0}\}$ at x .

The objective slope along the edge in direction $w^{(j)}$ is therefore, per unit of x_j ,

$$c^\top w^{(j)} = c_j - c_B^\top \mathbf{A}_B^{-1} \mathbf{A}e_j = \bar{c}_j, \quad \text{where } \bar{c}^\top := c^\top - c_B^\top \mathbf{A}_B^{-1} \mathbf{A}$$

is the vector of *reduced costs* (note $\bar{c}_B = \mathbf{0}$). Thus $\bar{c}_j$ is the true cost per unit increase of x_j when moving along the edge: the naive rate c_j , corrected by the term $-c_B^\top \mathbf{A}_B^{-1} \mathbf{A}e_j$ accounting for the forced adjustment of the basic variables. Compactly, eliminating the basic variables gives $c^\top x = c_B^\top \mathbf{A}_B^{-1}b + \bar{c}^\top x$ for every x with $\mathbf{Ax} = b$: the objective in the non-basic coordinates alone.

Algorithm 1: Simplex method, standard form, non-degenerate (one iteration)

Input: basis B with BFS x , $x_B = \mathbf{A}_B^{-1}b > \mathbf{0}$.
1 compute reduced costs $\bar{c}^\top = c^\top - c_B^\top \mathbf{A}_B^{-1} \mathbf{A}$;
2 **if** $\bar{c} \geq \mathbf{0}$ **then return** x is optimal;
3 choose an *entering* index j with $\bar{c}_j < 0$; // pivot rule chooses j
4 compute the edge direction: $w_j = 1$, $w_B = -\mathbf{A}_B^{-1} \mathbf{A} e_j$, w zero elsewhere;
5 **if** $w_B \geq \mathbf{0}$ **then return** *unbounded* along ray $x + t w$;
6 ratio test: $t^* \leftarrow \min_{i \in B: w_i < 0} x_i / (-w_i)$, attained at *leaving* index i^* ;
7 update $x \leftarrow x + t^* w$, $B \leftarrow B \cup \{j\} \setminus \{i^*\}$;

Algorithm 1 is the abstract simplex method, translated line by line through the dictionary. Line 2 is the optimality test of Theorem 4(a) (stated there for maximization; apply it to $-c$): $\bar{c} \geq \mathbf{0}$ says that every incident edge direction $w^{(j)}$ has slope $c^\top w^{(j)} = \bar{c}_j \geq 0$, so no edge improves the objective and x is optimal. Lines 3–4 select an improving edge and compute its direction. If $w_B \geq \mathbf{0}$ (line 5), then $x + t w \geq \mathbf{0}$ for all $t \geq 0$: the edge is an unbounded ray along which the objective decreases at rate $|\bar{c}_j|$. Otherwise, the ratio test of line 6 computes the largest step t^* keeping $x_B + t w_B \geq \mathbf{0}$ — geometrically, the other endpoint of the edge: the first basic variable to hit 0 is the *leaving* variable x_{i^*} , the relaxed bound $x_j = 0$ is exchanged for the newly tight bound $x_{i^*} = 0$, and the new point is the BFS of the neighboring basis $B \cup \{j\} \setminus \{i^*\}$. Each iteration costs one linear system solve (plus cheap updates in practice, where $\mathbf{A}_B^{-1}$ is maintained in factored form). Under non-degeneracy $t^* > 0$, so the objective strictly decreases by $t^* |\bar{c}_j|$ at every pivot and Theorem 5 applies verbatim.

Initialization: the two-phase scheme. The simplex method needs an initial BFS, but finding *any* feasible point is a priori as hard as LP itself. The resolution: feasibility is itself an LP with a readily available starting vertex. Flipping signs of rows so that $b \geq \mathbf{0}$, solve the *Phase 0* program

$$\min \mathbf{1}^\top z \text{ s.t. } \mathbf{A}x + z = b, \quad x, z \geq \mathbf{0},$$

in the variables $(x, z) \in \mathbb{R}^{n+m}$, starting from the BFS $(x, z) = (\mathbf{0}, b)$ with basis the z -columns. Its optimal value is 0 if and only if the original program is feasible, in which case an optimal BFS has $z = \mathbf{0}$ and yields a starting vertex for the real objective — *Phase 1* — after dropping the auxiliary variables. (If some auxiliary variable lingers in the basis at value 0, a degenerate case, it can be pivoted out; we skip these details.) Note that this feasibility LP is precisely the one behind the proof of the Farkas lemma in Lecture 1, now put to computational work.

3 Pivot Rules

A pivot rule decides which improving edge to take. This choice determines the behavior, and the complexity, of the simplex method.

Dantzig's rule. In standard form, Dantzig's original rule chooses the entering index j with the *most negative reduced cost* $\bar{c}_j$: the edge whose objective improvement per unit of the entering variable is largest. It is simple and cheap, though sensitive to the scaling of the variables. Other popular rules include *greatest improvement* (choose j maximizing the actual decrease $t^* |\bar{c}_j|$, requiring a ratio test per candidate), and *steepest edge* (choose the edge direction whose angle with $-c$

is smallest — empirically the strongest, and, in its approximate *Devex* variant, the default in most modern solvers).

The shadow vertex rule. The rule that will matter most for us is best understood visually. We stay in standard form and, for this discussion, consider a *pair* of objectives: an objective c whose unique minimizer v_0 we already know, and a *target* objective d that we wish to minimize. Now slide the objective from c to d along the segment

$$c_\lambda := (1 - \lambda)c + \lambda d, \quad \lambda : 0 \rightarrow 1,$$

and maintain, at every moment, a vertex minimizing c_λ . As λ grows, the optimal vertex changes at finitely many breakpoints, and consecutive optimal vertices are neighbors: the visited vertices form a simplex path from v_0 to an optimal vertex for d . Equivalently — and this explains the name — project the feasible region P onto the plane of the two objectives, $\pi(x) := (c^\top x, d^\top x)$. A vertex is visited exactly when it minimizes some nonnegative combination of c and d , i.e., when it projects onto the “lower-left” boundary of the two-dimensional *shadow* $\pi(P)$ — the *trade-off curve* recording the best d -value achievable for each c -value — traversed from the c -minimizing end to the d -minimizing end (Figure 4).

Where does the known objective c come from? When the rule serves as a pivot rule — optimize a given objective starting from a given vertex v_0 with basis B — the LP objective plays the role of d , and c is *constructed*: any c with $c_B = \mathbf{0}$ and $c_N > \mathbf{0}$ works, since then $c^\top x = c_N^\top x_N \geq 0$ for every feasible x , with equality only at $x_N = \mathbf{0}$, i.e., at v_0 (the tight-constraint construction from the proof of Theorem 6). Our naming, with the walk starting at the c -minimizer, is chosen to match the rule’s reappearances in Lectures 3 and 4.

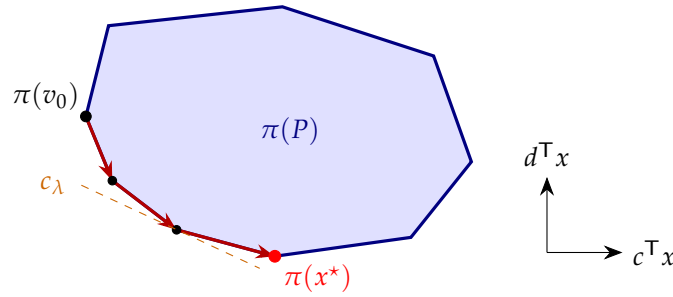


Figure 4: The shadow vertex rule: project the feasible region onto the plane of the two objectives, $\pi(x) = (c^\top x, d^\top x)$. The rule walks the lower-left boundary of the shadow $\pi(P)$ — the trade-off curve between minimizing c and minimizing d — from the c -optimal starting vertex $\pi(v_0)$ to the d -optimal vertex $\pi(x^*)$; each visited vertex minimizes some intermediate objective c_λ (dashed supporting line). The number of pivots is at most the number of vertices of the shadow polygon $\pi(P)$.

Under non-degeneracy, the implementation is a small tweak of Algorithm 1: maintain a basis B optimal for c_λ over a maximal interval $[\lambda_{\text{lo}}, \lambda_{\text{hi}}]$. The reduced costs $\bar{c}(\lambda) = (1 - \lambda)\bar{c} + \lambda\bar{d}$ are affine in λ , so $\lambda_{\text{hi}} = \max\{\lambda \leq 1 : \bar{c}(\lambda) \geq \mathbf{0}\}$ is computed by a ratio test over the coordinates; the entering index is the j whose reduced cost hits 0 at λ_{hi} , and the leaving index comes from the usual ratio test. Each pivot strictly increases λ_{hi} under non-degeneracy, and the walk ends when $\lambda_{\text{hi}} = 1$.

The shadow vertex rule is rarely used in practice — steepest edge outperforms it considerably — but it is particularly amenable to analysis: whether a vertex is visited depends only on the *data* (P, c, d) — namely, on whether it minimizes some intermediate objective c_λ — and not on the history of the walk. This *locality* is what has enabled the known probabilistic analyses of the simplex method, as we will see in later lectures, where the trade-off curve of Figure 4 will reappear several times — always traversed starting from the c -optimal end. The rule goes back to Gass and Saaty [GS55] under the name *parametric objective method*.

The random facet rule. The best worst-case guarantees come from *randomization*. Returning to the geometric form — $P = \{x : Ax \leq b\}$ of dimension n with m facets, maximizing $c^\top x$ — Kalai’s *random facet* rule is recursive. At the current vertex v : if no incident edge improves, stop. Otherwise, choose a facet F containing v *uniformly at random*, and recursively optimize over F — an LP of dimension $n - 1$ — starting from v . If the optimal vertex w of F is optimal for P , stop; otherwise, take an improving edge from w (it must exit F) and recurse from its other endpoint. Kalai [Kal92] and, independently and in dual form, Matoušek, Sharir and Welzl [MSW96] proved that the expected number of pivots is at most $e^{O(\sqrt{n \log m})}$ on *every* LP — a *subexponential* bound, $2^{O(\sqrt{n \log n})}$ when $m = \text{poly}(n)$. Thirty years on, this remains the state of the art for the worst-case complexity of the simplex method under any pivot rule; an improved variant of the rule appears in [HZ15], and nearly matching subexponential *lower* bounds for the rule itself were constructed by Friedmann, Hansen and Zwick [FHZ11]. The recursive, dimension-reducing style of its analysis will reappear in the Kalai–Kleitman diameter bound of Section 4.

Exponential examples: the Klee–Minty cube. None of these rules is known to be efficient in the worst case — and for most classical rules, explicit exponential examples are known. The prototype is the *Klee–Minty cube* [KM72]: for a parameter $0 < \varepsilon < 1/2$, consider

$$\begin{aligned} \max \quad & x_n \\ \text{s.t.} \quad & 0 \leq x_1 \leq 1, \\ & \varepsilon x_{j-1} \leq x_j \leq 1 - \varepsilon x_{j-1}, \quad j = 2, \dots, n. \end{aligned}$$

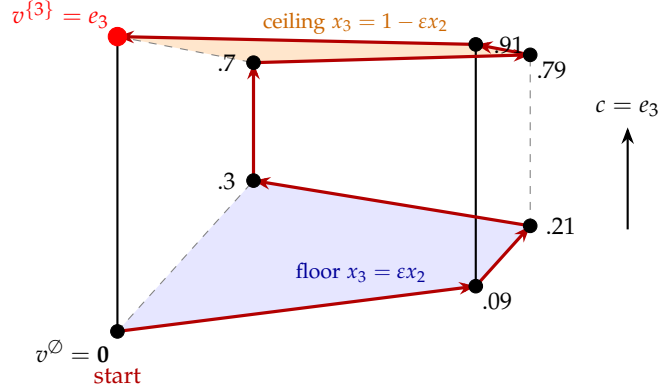


Figure 5: The Klee–Minty cube for $n = 3$, $\varepsilon = 0.3$, drawn with true vertex coordinates (height = objective x_3 ; each vertex is labeled by its x_3 -value). Starting at $v^\emptyset = 0$, the walk of Theorem 9 first traverses the floor facet $x_3 = \varepsilon x_2$ (blue), crosses to the ceiling facet $x_3 = 1 - \varepsilon x_2$ (orange), and traverses it in reverse, visiting all $2^3 = 8$ vertices with x_3 strictly increasing: $0 < .09 < .21 < .3 < .7 < .79 < .91 < 1$.

We write P_n for its feasible region, and refer to the constraints $\varepsilon x_{j-1} \leq x_j$ and $x_j \leq 1 - \varepsilon x_{j-1}$ as the *floor* and *ceiling* of the pair j (with floor $0 \leq x_1$ and ceiling $x_1 \leq 1$ for $j = 1$). The region P_n is a combinatorial cube: as we now verify, its vertices are indexed by the 2^n subsets $S \subseteq [n]$, recording for each pair whether the floor or the ceiling is tight.

LEMMA 8 (VERTICES OF THE KLEE–MINTY CUBE) *For each $S \subseteq [n]$, define $v^S \in \mathbb{R}^n$ recursively by*

$$v_j^S = \begin{cases} 1 - \varepsilon v_{j-1}^S, & j \in S, \\ \varepsilon v_{j-1}^S, & j \notin S, \end{cases} \quad j = 1, \dots, n \quad (\text{with } v_0^S := 0).$$

Then $V(P_n) = \{v^S : S \subseteq [n]\}$, all 2^n points being distinct with coordinates in $[0, 1]$, and at each v^S exactly one constraint of each pair j is tight: the ceiling if $j \in S$, the floor if $j \notin S$.

PROOF: First, every point $x \in P_n$ has $x_j \in [0, 1]$ for all j , by induction on j : given $x_{j-1} \in [0, 1]$, the pair- j constraints confine x_j to $[\varepsilon x_{j-1}, 1 - \varepsilon x_{j-1}] \subseteq [0, 1]$. Moreover, no point satisfies both constraints of a pair with equality: this would force $x_{j-1} = 1/(2\varepsilon) > 1$. Hence at most one constraint per pair is tight, so a vertex — which requires n tight constraints of rank n — makes *exactly one* constraint of each pair tight. Recording the set S of pairs whose ceiling is tight, forward substitution solves the tight system uniquely: its j -th row involves only x_{j-1} and x_j , with coefficient 1 on x_j , so the system is triangular with unit diagonal — in particular its rows are linearly independent of rank n — and its unique solution is v^S . Conversely, each v^S is feasible — the tight constraint of pair j holds with equality and the other one strictly, since floor and ceiling differ by $1 - 2\varepsilon x_{j-1} \geq 1 - 2\varepsilon > 0$ — and is therefore a vertex, being feasible with n tight linearly independent constraints. Distinctness follows since distinct sets S make distinct constraint sets tight, and no point makes more than n constraints tight. $\square$

The ε -tilting of the facets now orders these 2^n vertices so that a Hamiltonian path through *all* of them strictly improves the objective x_n at every step (Figure 5).

THEOREM 9 (KLEE–MINTY LOWER BOUND [KM72]) For $0 < \varepsilon < 1/2$, the vertex-edge graph of P_n admits a Hamiltonian path from $v^\varnothing = \mathbf{0}$ to $v^{\{n\}} = e_n$ along which x_n strictly increases. Consequently, starting from $\mathbf{0}$, the simplex method with objective $\max x_n$ can be made to perform $2^n - 1$ pivots, every one of them improving.

PROOF: We induct on n . For $n = 1$, the path is $0 \rightarrow 1$. Let $n \geq 2$, and let $w^1, \dots, w^{2^{n-1}}$ be the Hamiltonian path in P_{n-1} given by induction, along which x_{n-1} strictly increases from 0 (at $w^1 = v^\varnothing$) onwards.

By Theorem 8, every vertex of P_n makes exactly one constraint of pair n tight, and thus lies on exactly one of the two facets

$$F_{\text{fl}} := \{x \in P_n : x_n = \varepsilon x_{n-1}\}, \quad F_{\text{ce}} := \{x \in P_n : x_n = 1 - \varepsilon x_{n-1}\}.$$

Both facets are affine copies of P_{n-1} : the map $y \mapsto (y, \varepsilon y_{n-1})$ is an affine bijection from P_{n-1} onto F_{fl} (the pair- n constraints holding automatically, as $\varepsilon y_{n-1} \leq 1 - \varepsilon y_{n-1}$), and it identifies vertices with vertices and edges with edges, since it matches tight sets; similarly for $y \mapsto (y, 1 - \varepsilon y_{n-1})$ and F_{ce} . We claim the following is a path as required (writing $N := 2^{n-1}$ and $\mu := w_{n-1}^N$ for the largest value of x_{n-1}):

$$(w^1, \varepsilon w_{n-1}^1) \rightarrow \dots \rightarrow (w^N, \varepsilon \mu) \rightarrow (w^N, 1 - \varepsilon \mu) \rightarrow \dots \rightarrow (w^1, 1 - \varepsilon w_{n-1}^1).$$

It visits each vertex of F_{fl} and then each vertex of F_{ce} exactly once, i.e., all 2^n vertices of P_n . Its steps within the two facets are edges of P_n , being images of the edges of P_{n-1} under the two affine bijections. The middle step exchanges the floor of pair n for its ceiling while keeping the $n - 1$ tight constraints of w^N in P_{n-1} : the two endpoints thus share $n - 1$ tight constraints of rank $n - 1$, so the face of P_n these define is one-dimensional and bounded — it is exactly the segment between them, an edge of P_n (Lecture 1, faces are tight sets).

It remains to check that x_n strictly increases. On F_{fl} we have $x_n = \varepsilon x_{n-1}$ with x_{n-1} strictly increasing along $w^1, \dots, w^N$. The middle step increases x_n from $\varepsilon \mu$ to $1 - \varepsilon \mu$, since $\mu \leq 1 < 1/(2\varepsilon)$. On F_{ce} we have $x_n = 1 - \varepsilon x_{n-1}$ with x_{n-1} now strictly *decreasing* (the path runs in reverse), so x_n strictly increases. Finally, the path starts at $(w^1, 0) = \mathbf{0} = v^\varnothing$ and ends at $(w^1, 1 - 0) = e_n = v^{\{n\}}$. $\square$

Theorem 9 exhibits a strictly improving simplex path of length $2^n - 1$; an adversarially chosen improving pivot rule therefore requires exponentially many iterations, already on a polytope with $2n$ facets. The stronger statement — that *specific* rules are forced onto this path — requires calibrating the instance to the rule, so that the reduced cost of the “wrong” entering variable always looks most attractive. (Some calibration is genuinely needed: on the instance as stated above, Dantzig’s rule started at $\mathbf{0}$ reaches the optimum in a *single* pivot.) For Dantzig’s rule, the calibration is the recursive change of variables

$$x_j = \varepsilon x_{j-1} + \theta^{j-1} \tilde{x}_j, \quad j = 1, \dots, n,$$

for a second small parameter $0 < \theta < \varepsilon$. In the new variables, Dantzig’s rule follows the Hamiltonian path of Theorem 9 exactly; for $\varepsilon = 1/3$, $\theta = 1/9$, the resulting instance is precisely the classical formulation of Klee and Minty [KM72]. Similar constructions exist for virtually every deterministic pivot rule proposed before the 2000s, including steepest edge and the shadow vertex rule (for the latter, via the *Goldfarb cube*, Exercise 2). Randomized and history-dependent rules

resisted longer, but sub-exponential lower bounds are now known for the natural candidates as well. Whether *some* pivot rule runs requires only a polynomial number of pivots is a central open question of the field.

Exercise 2 (Goldfarb cube) Let $0 < \beta < \alpha/4 < 1/8$, and consider $\max x_n$ subject to $0 \leq x_1 \leq 1$, $\alpha x_1 \leq x_2 \leq 1 - \alpha x_1$, and

$$\alpha(x_{k-1} - \beta x_{k-2}) \leq x_k \leq 1 - \alpha(x_{k-1} - \beta x_{k-2}), \quad 3 \leq k \leq n.$$

(Here it is convenient to run the shadow vertex rule in maximization form; equivalently, negate both objectives.)

1. Show that the feasible region is a combinatorial cube: at every vertex exactly one constraint of each pair is tight, and the 2^n vertices are indexed by the sets $S \subseteq [n]$ of pairs whose upper constraint is tight. (Hint: Following the proof of Theorem 8, prove by induction on k that every feasible point satisfies $x_k \geq \frac{\alpha}{2} x_{k-1} \geq 0$ and $x_k \leq 1$. Conclude that both constraints of a pair are never tight simultaneously, and that the tight systems are triangular.)
2. Show that every vertex is the unique maximizer of some objective $\gamma e_{n-1} + \delta e_n$. (Hint: By Lecture 1, it suffices to write $\gamma e_{n-1} + \delta e_n$ as a positive combination $\sum_k y_k a^{(k)}$ of the n tight constraint normals $a^{(k)} = \pm e_k + \alpha e_{k-1} - \alpha\beta e_{k-2}$ (with $e_0 = e_{-1} := \mathbf{0}$). Encode the weights by the ratios $s_k := y_k/y_{k+1}$: cancelling coordinate $k \leq n-2$ determines s_{k+1} from s_k . Choose $s_1 \in (0, \alpha/2]$ and propagate backwards, showing — using $\beta < \alpha/4$ — that every ratio required along the way stays in $(0, \alpha/2]$; finally unwind the weights from $y_n := 1$.)
3. Conclude that the shadow vertex rule interpolating towards e_n can be forced to visit all 2^n vertices. (Hint: By locality, a vertex is visited as soon as it uniquely maximizes some intermediate objective; choose the starting objective so that all the objectives (γ, δ) produced in part 2 occur along the interpolation.)

4 The Diameter of Polyhedra

For this section we return to the geometric form $P = \{x : \mathbf{A}x \leq b\}$ of Lecture 1: the diameter is a statement about facets, and the facet count m is its natural parameter. (Nothing algorithmic happens here; the computational form returns in the next lecture.) Throughout, we write

$$I(v) := \{i \in [m] : a_i^T v = b_i\}$$

for the set of constraints tight at a vertex v , as in Theorem 2 — geometrically, assuming a non-redundant description, $I(v)$ is the set of facets of P containing v .

For any pivot rule, the number of pivots needed to move between two vertices is at least their distance in the graph $G(P)$. The following quantity therefore yields a worst-case lower bound: the (*combinatorial*) *diameter*

$$\text{diam}(P) := \max_{u, v \in V(P)} \text{dist}_{G(P)}(u, v),$$

the largest graph distance between two vertices. If some family of polytopes had super-polynomial diameter, then *no* pivot rule whatsoever could make the simplex method polynomial on them. Remarkably, we do not know whether this happens.

Polynomial Hirsch conjecture (open). Is $\text{diam}(P)$ bounded by a polynomial in n and m for every n -dimensional polyhedron P with m facets?

Hirsch conjectured in 1957 (in a letter to Dantzig [Dan63]) the elegant bound $\text{diam}(P) \leq m - n$. For polytopes this stood for over fifty years, until Santos [San12] constructed a counterexample; even so, all known lower bounds are within a constant factor of $m - n$, while the best general upper bound, due to Kalai and Kleitman [KK92], is quasi-polynomial. Proving their bound is our goal for this section.

THEOREM 10 (KALAI–KLEITMAN) *Every n -dimensional polytope P with m facets satisfies $\text{diam}(P) \leq m^{1+\log_2 n}$.*

What is remarkable about the proof is how little geometry it uses: it works with vertices purely as *sets of facet labels*, plus one connectivity property. We first explain in what sense the diameter is a purely combinatorial quantity.

4.1 Vertices as sets of facets: the face lattice

Recall from Lecture 1 that the faces of P are its subsets of the form $\{x \in P : \mathbf{A}_I x = b_I\}$ for tight sets I , ranging from $\emptyset$ and the vertices up to the facets and P itself. Ordered by inclusion, they form the *face lattice* $\mathcal{L}(P)$ (Figure 6): the intersection of two faces is again a face (their *meet*), and the smallest face containing a given pair is their *join*. The face lattice records every combinatorial feature of P — which vertices lie on which facets, which pairs are joined by edges — while forgetting all metric information; polytopes with isomorphic face lattices are called *combinatorially equivalent*. For instance, every cube-like P_n of Theorem 8 is combinatorially equivalent to the cube $[0, 1]^n$.

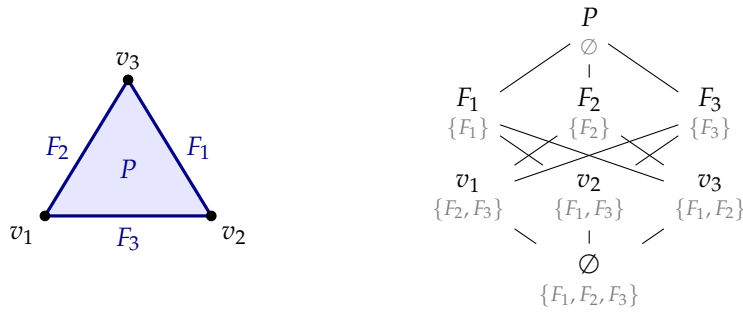


Figure 6: A triangle ($n = 2$, $m = 3$) and its face lattice, drawn as a Hasse diagram: each face is joined to the faces immediately above and below it in the inclusion order, and is labeled (gray) with its set of tight constraints — none for P itself, all three for the empty face. Note that inclusion of faces *reverses* inclusion of tight sets. Each vertex v_i lies on exactly $n = 2$ facets, so it is encoded by its 2-element tight set $I(v_i)$; e.g. $I(v_1) = \{2, 3\}$, the labels of F_2 and F_3 .

We assume for the rest of the section that P is *simple*: every vertex lies on exactly n facets. This is without loss of generality for diameter upper bounds: the perturbation $b \mapsto b + (\varepsilon, \varepsilon^2, \dots, \varepsilon^m)^\top$ of Exercise 1 makes P simple for small $\varepsilon > 0$, and it can only increase the diameter — each vertex of P splits into a nonempty cluster of vertices of the perturbed polytope, and paths of

the perturbed graph project back onto walks of $G(P)$; we omit the verification. For a simple polytope, the encoding of vertices by facet sets is particularly clean: the map $v \mapsto I(v)$ is injective into $\binom{[m]}{n}$, two vertices u, v are adjacent in $G(P)$ exactly when $|I(u) \cap I(v)| = n - 1$, and, more generally, the faces containing v correspond to the subsets of $I(v)$ (the face with tight set $J \subseteq I(v)$ being $(n - |J|)$ -dimensional). The diameter of P is thus entirely determined by the set family $\{I(v) : v \in V(P)\}$ — an invitation to discard the geometry altogether and bound the diameter of *abstract* families of n -element sets. Following Eisenbrand, Hähnle, Razborov and Rothvoß [EHRR10], we distill the one geometric property the argument needs — connectivity of the graphs of faces — into the following object.

4.2 Connected layer families

DEFINITION 11 (CONNECTED LAYER FAMILY) *Let $d, N \geq 1$. A connected layer family (CLF) of order d on a ground set of N symbols is a sequence $L_1, \dots, L_\ell$ of nonempty collections of d -element subsets of the ground set, with all sets appearing in the family distinct, satisfying the connectivity property: for all $i \leq j \leq k$ and all $A \in L_i, C \in L_k$, there exists $B \in L_j$ with $B \supseteq A \cap C$. The length of the family is the number ℓ of layers.*

All the arguments below rest on two closure properties of the definition; proving them is a good warm-up with the connectivity property (Exercise 3).

LEMMA 12 (CLF CALCULUS) *Let $L_1, \dots, L_\ell$ be a CLF of order d on N symbols.*

- (i) (Taking sublayers.) *For all indices $i_1 < i_2 < \dots < i_r$, the subsequence $L_{i_1}, \dots, L_{i_r}$ is a CLF of order d .*
- (ii) (Restricting to a common set.) *Let S be a set of at most $d - 1$ symbols. The layers containing at least one set $\supseteq S$ form a contiguous (possibly empty) interval of $\{1, \dots, \ell\}$. Restricting these layers to their sets containing S , and deleting the symbols of S from each, yields a CLF of order $d - |S|$ on at most $N - |S|$ symbols, of the same length as the interval.*

In particular, taking $S = \{s\}$ in (ii): each symbol s occupies a consecutive interval of layers. Figure 7 shows this interval structure on a small example.

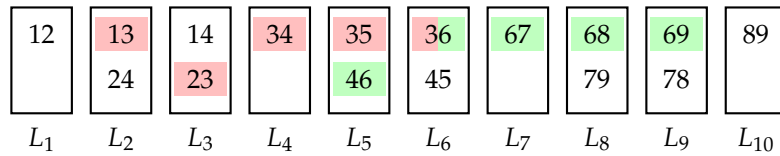


Figure 7: A connected layer family of order $d = 2$ on $N = 9$ symbols, of length 10 (each 2-set $\{a, b\}$ is abbreviated ab). Each symbol occupies a contiguous interval of layers: the sets containing symbol 3 (red) span layers 2 through 6, those containing symbol 6 (green) span layers 5 through 9, and the set 36 in layer 6 belongs to both intervals. Here the interval property is exactly the connectivity property, since two sets in distinct layers intersect in at most one symbol.

The geometric source of such families is graph distance on a polytope:

LEMMA 13 (POLYTOPES YIELD CLFs) *Let P be a simple n -polytope with m facets, let $u \in V(P)$, and identify each vertex v with its n -element tight set $I(v)$. Then*

$$L_i := \{I(v) : \text{dist}_{G(P)}(u, v) = i\}, \quad i = 0, 1, \dots, \ell - 1,$$

(discarding empty tail layers) is a CLF of order n on m symbols, of length $\ell = 1 + \max_v \text{dist}(u, v)$.

PROOF: For a simple polytope, $v \mapsto I(v)$ is injective (v is the unique point on all n of its facets), so the sets are distinct; the layers are nonempty for i up to the eccentricity of u because distances change by at most 1 along edges. For connectivity, let $A = I(v)$, $C = I(v')$ with $\text{dist}(u, v) = i$, $\text{dist}(u, v') = k$, $i \leq j \leq k$, and let $F := \{x \in P : \text{constraints } A \cap C \text{ tight}\}$, a face of P containing v and v' . The graph of F is connected — F is itself a polytope, so Theorem 6 applies to it — so there is a path from v to v' inside F (edges of F are edges of P , faces of faces being faces). Along this path, the distance to u starts at i , ends at k , and changes by at most 1 per step, so some vertex w on it has $\text{dist}(u, w) = j$; its set $B = I(w)$ contains $A \cap C$ since $w \in F$. $\square$

4.3 The Kalai–Kleitman bound

By Theorem 13 (choosing u in a diameter-realizing pair), Theorem 10 follows immediately from a purely combinatorial statement:

THEOREM 14 (ABSTRACT KALAI–KLEITMAN) *Every CLF of order d on $N \geq 2$ symbols has length $\ell \leq N^{1+\log_2 d}$.*

PROOF: For $d \geq 1$ and $M \geq 0$, let $h(d, M)$ denote the maximum length of a CLF of order d on a ground set of M symbols, with $h(d, M) := 0$ if no such family exists — in particular, $h(d, M) = 0$ whenever $M < d$, since a nonempty layer needs a d -element subset of the ground set. Note that h is non-decreasing in M , and that $h(1, M) \leq M$, since the sets of an order-1 family are distinct singletons. The heart of the proof is the recursion

$$h(d, M) \leq 2h(d, \lfloor M/2 \rfloor) + h(d-1, M-1), \quad d \geq 2. \quad (1)$$

To prove it, consider a CLF $L_1, \dots, L_\ell$ of order d on M symbols (Figure 8). Let t be the largest index such that the layers $L_1, \dots, L_t$ together use at most $\lfloor M/2 \rfloor$ symbols (possibly $t = 0$). These layers form a CLF of order d on at most $\lfloor M/2 \rfloor$ symbols (Theorem 12(i)), so $t \leq h(d, \lfloor M/2 \rfloor)$, by the very definition of h — this is why we work with the maximum length function rather than the bound we are trying to prove. Symmetrically, if t' is largest such that the *last* t' layers use at most $\lfloor M/2 \rfloor$ symbols, then $t' \leq h(d, \lfloor M/2 \rfloor)$. If $t + t' \geq \ell$ we are done, so assume otherwise. By maximality of t and t' , the prefix $L_1, \dots, L_{t+1}$ and the suffix $L_{\ell-t'}, \dots, L_\ell$ each use *more* than $\lfloor M/2 \rfloor$ symbols, so they share a symbol s : say $s \in A \in L_i$ with $i \leq t+1$ and $s \in C \in L_k$ with $k \geq \ell - t'$. By Theorem 12(ii) applied to $S = \{s\}$, the layers containing a set through s form an interval — which contains i and k , hence all of $\{i, \dots, k\}$ — and restricting them to the sets containing s and deleting s yields a CLF of order $d-1$ on at most $M-1$ symbols, of length at least $k - i + 1 \geq \ell - t' - (t+1) + 1$. Hence $\ell - t - t' \leq h(d-1, M-1)$, proving (1).

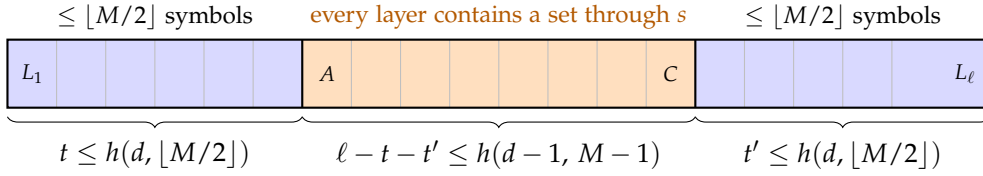


Figure 8: The Kalai–Kleitman recursion (1). The longest prefix using at most $\lfloor M/2 \rfloor$ symbols (blue, left) has length $t \leq h(d, \lfloor M/2 \rfloor)$; symmetrically for the suffix (blue, right). If the family is longer than $t + t'$, the extended prefix $L_1, \dots, L_{t+1}$ and suffix $L_{\ell-t'}, \dots, L_\ell$ each use more than $\lfloor M/2 \rfloor$ symbols, hence share a symbol s : say $s \in A \in L_{t+1}$ and $s \in C \in L_{\ell-t'}$. By connectivity, every layer in between contains a set through s (orange); restricting these layers to the sets through s and deleting s leaves a CLF of order $d - 1$ on at most $M - 1$ symbols, so the middle stretch has length at most $h(d - 1, M - 1)$.

It remains to solve the recursion. Since $N^{1+\log_2 d} = (2d)^{\log_2 N}$, we show

$$h(d, N) \leq (2d)^{\log_2 N} \quad \text{for all } d \geq 1, N \geq 2,$$

by induction on N , for all orders d simultaneously. For $d = 1$ we have $h(1, N) \leq N = 2^{\log_2 N}$, and in the base case $N = 2$ there are at most $\binom{2}{d} \leq 2$ distinct d -element subsets, so $h(d, 2) \leq 2 \leq 2d$. For $N \geq 3$ and $d \geq 2$, we may also assume $d \leq N$ — otherwise $h(d, N) = 0$ and there is nothing to prove. Now apply (1) to its own last term repeatedly — using that h is non-decreasing in the number of symbols, $h(d - j, \lfloor (N - j)/2 \rfloor) \leq h(d - j, \lfloor N/2 \rfloor)$ — until the order reaches 1 (as $d \leq N$, every term encountered has a ground set of at least $N - d + 1 \geq 1$ symbols):

$$h(d, N) \leq 2 \sum_{i=2}^d h(i, \lfloor N/2 \rfloor) + h(1, N - d + 1) \leq 2 \sum_{i=2}^d h(i, \lfloor N/2 \rfloor) + N.$$

By the induction hypothesis at $\lfloor N/2 \rfloor < N$, each summand satisfies $h(i, \lfloor N/2 \rfloor) \leq (2i)^{\log_2(N/2)} \leq (2d)^{\log_2 N - 1}$ (summands with $\lfloor N/2 \rfloor < i$ are zero), while $N = 2 \cdot 2^{\log_2 N - 1} \leq 2(2d)^{\log_2 N - 1}$. Therefore

$$h(d, N) \leq 2(d - 1)(2d)^{\log_2 N - 1} + 2(2d)^{\log_2 N - 1} = (2d)^{\log_2 N}.$$

□

The exponent has been trimmed — Todd [Tod14] refined the same argument to $(m - n)^{\log_2 n}$ — but decades of effort have not moved the general upper bound below quasi-polynomial.

4.4 How long can a connected layer family be?

The Kalai–Kleitman recursion halves the number of *symbols*; an older argument of Larman [Lar70], refined by Barnette [Bar74], instead reduces the *order* at a cost of a factor 2, yielding a bound that is linear in m — far better than Theorem 10 in fixed dimension.

THEOREM 15 (ABSTRACT BARNETTE–LARMAN) *Every CLF of order d on N symbols has length at most $2^{d-1}N$. Consequently, every n -dimensional polytope with m facets satisfies $\text{diam}(P) \leq 2^{n-1}m$*

The proof, a greedy induction on the order d , is outlined in Exercise 4.

How far can the combinatorial abstraction be pushed — how long can a CLF actually be? Intriguingly, despite the definition retaining almost none of the geometry, *no super-polynomial lower bound is known*: the longest known constructions, due to Eisenbrand, Hähnle, Razborov and Rothvoß [EHRR10], have length $\Omega(N^2 / \log N)$ (with order $d = \Theta(N)$) — roughly Nd , and only mildly super-linear. Hähnle has conjectured that this is essentially the truth:

Hähnle’s conjecture (open). Every connected layer family of order d on N symbols has length at most $d(N - 1) + 1$.

The conjecture, posed during the *Polymath 3* project on the polynomial Hirsch conjecture [Kal10], is known to hold in two extremal cases — when every layer consists of a single set, and when *all* of $\binom{[N]}{d}$ appears — and the bound $d(N - 1) + 1$ is attained exactly in a relaxed model where layers consist of d -element *multisets*. Even the first non-trivial case $d = 3$ is open: the best upper bound there is $4N - 1$, against the conjectured $3N - 2$. If true, the conjecture would imply the linear diameter bound $\text{diam}(P) \leq n(m - 1)$ — dramatically stronger than the polynomial Hirsch conjecture. The overall state of affairs is thus: within the abstraction, the truth lies between $\Omega(N^2 / \log N)$ and $N^{1+\log_2 d}$, and either the upper bound can be improved dramatically, or CLFs of super-quasipolynomial length exist, and polynomial diameter bounds for polytopes must then use genuinely geometric arguments beyond the connectivity of faces. Both directions are wide open.

Exercise 3 (CLF calculus) Prove Theorem 12. (Hint: For (ii): if $A \supseteq S$ appears in layer i and $C \supseteq S$ in layer k , then for every $i \leq j \leq k$ connectivity provides $B \in L_j$ with $B \supseteq A \cap C \supseteq S$. Deleting the symbols of S keeps the sets distinct, since the sets involved all contain S .)

Exercise 4 (Proof of the Barnette–Larman bound) Let $L_1, \dots, L_\ell$ be a CLF of order $d \geq 2$ on N symbols, and let $h(d, N)$ be the maximum length of such a family, as in the proof of Theorem 14 (recall $h(1, N) \leq N$). For each symbol s , let $J_s \subseteq \{1, \dots, \ell\}$ be the interval of layers containing a set through s (Theorem 12(ii)).

1. Greedily partition the layers into intervals: having covered $L_1, \dots, L_t$, choose the symbol s occurring in layer L_{t+1} whose interval J_s extends furthest to the right, and let the next interval be $\{t + 1, \dots, \max J_s\}$. Conclude from Theorem 12(ii) that each interval produced has length at most $h(d - 1, N_k)$, where N_k is the number of symbols occurring in its layers.
2. Show that each symbol occurs in the layers of at most *two* consecutive intervals of the greedy partition. (Hint: Contiguity and the greedy choice.)
3. Conclude that $\sum_k N_k \leq 2N$, and by induction on d that $\ell \leq 2^{d-1}N$.

Summary and Next Lecture. In this lecture, we developed the simplex method, proved lower bounds on its complexity, and studied the graph structure of polyhedra. We showed that some pivot rules require exponentially many iterations in the worst case, and that even the existence of short vertex paths — the polynomial Hirsch conjecture — is wide open; yet simplex is widely used in practice thanks to its strong empirical performance. In the next lecture, we bridge this gap using *smoothed analysis*, which shows that worst-case instances such as the Klee–Minty cube are destroyed by small random perturbations of the data. For further reading on this lecture’s material, see the survey by Shamir [Sha87] for the practical performance of simplex, Ziegler [Zie95]

for polytope combinatorics, and Santos’s survey [San12] and the Polymath 3 project [Kal10] for the diameter question.

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